

# Symbiotic model - 11.21.25

## Astrangia poculata sym

COMPLETE = 3.0; MRE = 0.077; SMAE = 0.109; SMSE = 0.040

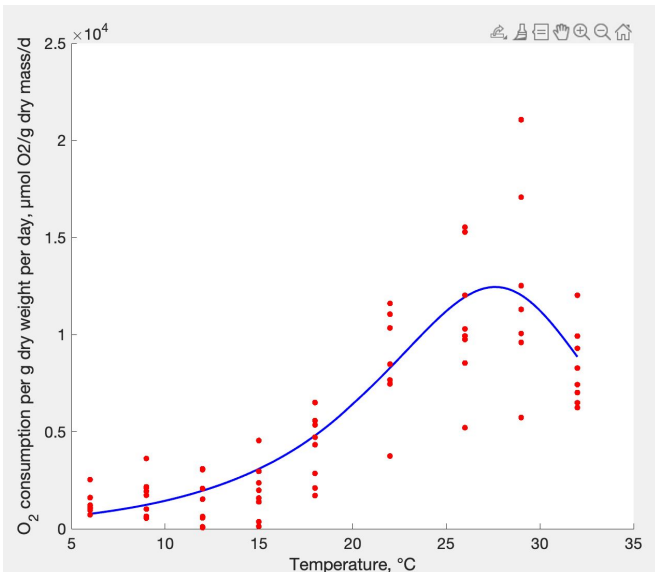
### Data & predictions

| data       | prd      | RE       | symbol      | units                                                   | description                     |
|------------|----------|----------|-------------|---------------------------------------------------------|---------------------------------|
| 1096       | 1094     | 0.001698 | tp          | d                                                       | time since birth at puberty     |
| 0.011      | 0.01102  | 0.001846 | L0          | cm                                                      | egg diameter                    |
| 0.0137     | 0.01365  | 0.003534 | Lb          | cm                                                      | total length at birth           |
| 0.8        | 0.7619   | 0.04762  | RQ          | -                                                       | Respiratory quotient            |
| see figure | 0.3051   | TJO      | Temperature | vs. O <sub>2</sub> consumption per g dry weight per day |                                 |
| 0.02       | 0.02393  | 0.1966   | psd.v       | cm/d                                                    | energy conductance              |
| 18         | 73.39    | 3.077    | psd.p_M     | J/d.cm <sup>3</sup>                                     | vol-spec som maint              |
| 0.002      | 0.002    | 0        | psd.k_J     | 1/d                                                     | maturity maint rate coefficient |
| 0.3        | 0.007122 | 0.9763   | psd.k       | -                                                       | maintenance ratio               |
| 0.8        | 0.7792   | 0.02603  | psd.kap     | -                                                       | allocation fraction to soma     |
| 0.8        | 0.8005   | 0.000653 | psd.kap_G   | -                                                       | growth efficiency               |
| 0.95       | 0.95     | 0        | psd.kap_R   | -                                                       | reproduction efficiency         |



### std parameters at 20.0°C

| symbol | units               | value     | free | description                               |
|--------|---------------------|-----------|------|-------------------------------------------|
| T_ref  | K                   | 293.1     | 0    | Reference temperature                     |
| T_A    | K                   | 1.239e+04 | 1    | Arrhenius temperature                     |
| z      | -                   | 0.06714   | 1    | zoom factor                               |
| F_m    | l/d.cm <sup>2</sup> | 6.5       | 0    | {F_m}, max spec searching rate            |
| kap_X  | -                   | 0.8       | 0    | digestion efficiency of food to reserve   |
| kap_P  | -                   | 0.1       | 0    | faecation efficiency of food to faeces    |
| v      | cm/d                | 0.02393   | 1    | energy conductance                        |
| kap    | -                   | 0.7792    | 1    | allocation fraction to soma               |
| kap_R  | -                   | 0.95      | 0    | reproduction efficiency                   |
| p_M    | J/d.cm <sup>3</sup> | 73.39     | 1    | {p_M}, vol-spec somatic maint             |
| p_T    | J/d.cm <sup>2</sup> | 0         | 0    | {p_T}, surf-spec somatic maint            |
| k_J    | 1/d                 | 0.002     | 0    | maturity maintenance rate coefficient     |
| E_G    | J/cm <sup>3</sup>   | 261.3     | 1    | {E_G}, spec cost for structure            |
| E_Hb   | J                   | 2.065e-05 | 1    | maturity at birth                         |
| E_Hp   | J                   | 1.571     | 1    | maturity at puberty                       |
| h_a    | 1/d <sup>2</sup>    | 3.158e-15 | 1    | Weibull aging acceleration                |
| s_G    | -                   | 0.0001    | 0    | Gompertz stress coefficient               |
| T_AH   | K                   | 3.526e+04 | 1    | Arrhenius temperature for upper boundary  |
| T_H    | K                   | 302.4     | 0    | upper boundary tolerance range            |
| del_M  | -                   | 0.4652    | 1    | shape coefficient for larvae              |
| del_W  | -                   | 1         | 0    | dry - ash free dry weight ratio           |
| f      | -                   | 1         | 0    | scaled functional response for 0-var data |



# Aposymbiotic model - 11.21.25

## Astrangia poculata apo

COMPLETE = 3.0; MRE = 0.091; SMAE = 0.139; SMSE = 0.039

### Data & predictions

| data       | prd      | RE       | symbol      | units                                                   | description                     |
|------------|----------|----------|-------------|---------------------------------------------------------|---------------------------------|
| 1096       | 1098     | 0.001897 | tp          | d                                                       | time since birth at puberty     |
| 0.011      | 0.01138  | 0.03478  | L0          | cm                                                      | egg diameter                    |
| 0.0137     | 0.01404  | 0.02479  | Lb          | cm                                                      | total length at birth           |
| 0.8        | 0.7619   | 0.04762  | RQ          | -                                                       | Respiratory quotient            |
| see figure | 0.3098   | TJO      | Temperature | vs. O <sub>2</sub> consumption per g dry weight per day |                                 |
| 0.02       | 0.04305  | 1.152    | psd.v       | cm/d                                                    | energy conductance              |
| 18         | 168.9    | 8.385    | psd.p_M     | J/d.cm <sup>3</sup>                                     | vol-spec som maint              |
| 0.002      | 0.002    | 0        | psd.k_J     | 1/d                                                     | maturity maint rate coefficient |
| 0.3        | 0.003076 | 0.9897   | psd.k       | -                                                       | maintenance ratio               |
| 0.8        | 0.7077   | 0.1154   | psd.kap     | -                                                       | allocation fraction to soma     |
| 0.8        | 0.8051   | 0.006434 | psd.kap_G   | -                                                       | growth efficiency               |
| 0.95       | 0.95     | 0        | psd.kap_R   | -                                                       | reproduction efficiency         |

### std parameters at 20.0°C

| symbol | units               | value     | free | description                               |
|--------|---------------------|-----------|------|-------------------------------------------|
| T_ref  | K                   | 293.1     | 0    | Reference temperature                     |
| T_A    | K                   | 1.636e+04 | 1    | Arrhenius temperature                     |
| z      | -                   | 0.04298   | 1    | zoom factor                               |
| F_m    | l/d.cm <sup>2</sup> | 6.5       | 0    | {F_m}, max spec searching rate            |
| kap_X  | -                   | 0.8       | 0    | digestion efficiency of food to reserve   |
| kap_P  | -                   | 0.1       | 0    | faecation efficiency of food to faeces    |
| v      | cm/d                | 0.04305   | 1    | energy conductance                        |
| kap    | -                   | 0.7077    | 1    | allocation fraction to soma               |
| kap_R  | -                   | 0.95      | 0    | reproduction efficiency                   |
| p_M    | J/d.cm <sup>3</sup> | 168.9     | 1    | [p_M], vol-spec somatic maint             |
| p_T    | J/d.cm <sup>2</sup> | 0         | 0    | {p_T}, surf-spec somatic maint            |
| k_J    | 1/d                 | 0.002     | 0    | maturity maintenance rate coefficient     |
| E_G    | J/cm <sup>3</sup>   | 259.8     | 1    | [E_G], spec cost for structure            |
| E_Hb   | J                   | 3.294e-05 | 1    | maturity at birth                         |
| E_Hp   | J                   | 1.903     | 1    | maturity at puberty                       |
| h_a    | 1/d <sup>2</sup>    | 3.598e-16 | 1    | Weibull aging acceleration                |
| s_G    | -                   | 0.0001    | 0    | Gompertz stress coefficient               |
| T_AH   | K                   | 3.862e+04 | 1    | Arrhenius temperature for upper boundary  |
| T_H    | K                   | 300.4     | 0    | upper boundary tolerance range            |
| del_W  | -                   | 1         | 0    | dry - ash free dry weight ratio           |
| del_M  | -                   | 0.4619    | 1    | shape coefficient for larvae and egg      |
| f      | -                   | 1         | 0    | scaled functional response for 0-var data |

